

SCI-JINC

Engineering Conformance Specification

Contract v0.1 / document r0.1

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Status. Implementation-blind Stage B author draft. This document expresses the same shared scientific authority as the companion rationale. It contains no implementation mapping and assesses no implementation candidate. A compiling specification is not evidence of conformity, representation fidelity, validation, achieved performance, readiness, or production status.

Contract identity	SCI-JINC v0.1
Document identity	Engineering Conformance Specification r0.1
Companion view	Scientific Rationale and Contract v0.1/r0.1
Normative core	The six shared modules under <code>src/common/</code> included verbatim in both views
Requirements	SCI-JINC-REQ-001 through SCI-JINC-REQ-042, sequential with no gaps
Predictions	SCI-JINC-PRED-001 through SCI-JINC-PRED-032, sequential with no gaps
Present assessment	Not assessed ; no implementation candidate was inspected

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1 Purpose and normative use

This specification turns the shared SCI-JINC scientific authority into a future conformance-review surface. It does not introduce independent science. If prose in this engineering introduction appears broader than the shared definitions, equations, assumptions, requirements, or predictions, the shared module controls and the broader engineering reading is invalid.

The conformance object is one claimed realization of the SCI-JINC observation estimator under one exact contract version, parameter-set identity, coefficient-family identity, target WCS, array, observation, and lifecycle generation. A future assessment must bind that object before drawing any conclusion. This author draft binds no implementation, schema, interface, file, function, algorithm, or execution.

Conformance is evaluated requirement by requirement. Each result is exactly one of *pass*, *fail*, *not assessed*, or *not applicable under a contract-authorized condition*. Missing evidence is *not assessed*, never pass. A pass at one layer does not promote a claim at another layer.

2 Applicability and fail-closed preconditions

A numerical conformance route exists only when all scientific identities and inputs required by the shared core are available. In particular:

1. the exact transformed signal occurrence and exact same-processed-sample AST coordinate association are available;
2. one exact PTC coefficient family/version explicitly permits SCI-JINC, is selected under the authorized lifecycle, and supplies compatible realized payload and QC;
3. one exact scientifically authorized TolTEC array-associated kernel parameter set is resolved; and
4. target WCS, pixel size, effective `subpixel_n`, processed-sample frequency, observation, array, plan, destination, and lifecycle identities are complete and mutually compatible.

Absence of a coefficient route or TolTEC parameter set is a typed unavailable state. It cannot be converted into an executable route by unity, a MAP family, an alternate family, an inherited 45 m denominator, inherited shape values, mode-dependent $r_{\max}$, or any other hidden implementation default. A future assessment may test fail-closed behavior without claiming a scientifically authorized numerical production route.

3 Future conformance procedure

A later authorized assessor should proceed in the following order:

1. **Bind identity.** Record the exact SCI-JINC contract, candidate, observation, stable array, plan/realization, target WCS, coefficient family, parameter set, destination, and lifecycle generation.
2. **Check source availability.** Confirm that every producer-owned fact is present under the exact boundary and that no excluded fallback or ordinary MAP inheritance is used.
3. **Check algebra and membership.** Establish the exact occurrence population, AST association, center-domain decision, square membership, point-phase coefficient, and coupled accumulator equations.
4. **Check units and limits.** Exercise the zero limit, signed lobes, coefficient classes, scaling predictions, cancellation states, and coefficient-squared-time unit.
5. **Check grouping and publication.** Exercise array cardinality, destination uniqueness, same-observation chunking, bundle atomicity, and absence of unauthorized roles or coaddition.
6. **Disclose numerical policy.** State the function-evaluation, phase/cache, accumulation/reduction, and comparison policies and demonstrate combined error negligible relative to the approximately 10^{-3} scale.
7. **Run every prediction.** Record construction, expected result, observed result, tolerance rationale, and pass/fail/not-assessed state for each stable prediction identifier.
8. **Separate conclusions.** Report algebraic conformance separately from representation fidelity, numerical and observational validation, achieved performance, readiness, and production authorization.

This sequence is not an implementation architecture. It is an evidence ordering intended to prevent a successful compile, one numerical example, or one stored product from being mistaken for complete scientific conformance.

4 Minimum future evidence record

Table 1: Evidence categories for a later authorized assessment.

Category	Minimum record	Related requirements
Authority and identity	Exact contract/candidate identifiers; observation, array, plan, WCS, coefficient family, parameter set, destination, and lifecycle bindings.	001–007, 030–035
Producer boundaries	Exact PTC signal/coefficient/QC and AST coordinate association with immutable parentage and dispositions.	002–006, 035–037
Analytic operator	Zero limit, parameter roles, point phase, square extent, center gate, crop, and finite signed coefficient behavior.	007–015
Accumulator algebra	Exact membership and separate N_p , C_p , Q_p , m_p , A_{pi} , and $T_p^{(\kappa^2)}$ computations.	016–020, 026–027
Conditioning and numerical adequacy	D_p , ρ_p , error budget, comparison policy, and accepted/rejected cancellation constructions.	019–023
Units and statistical claims	Unit table, scale transformations, coefficient meaning, and proof that conditional formulas are not promoted.	024–025, 038
Bundle and lifecycle	Five-role schema, local versus whole failure, zero-through-three arrays, unique destination, same-observation chunking, and no cross-observation route.	028–034
Excluded products and defaults	Demonstration that no forbidden fallback, inherited parameter, extra role, diagnostic/provenance framework, or MAP rule is silently introduced.	005, 038–041
Prediction results	Complete result for each SCI-JINC-PRED-NNN, including disclosed tolerance and evidence locator.	042 and prediction trace
Claim register	Separate states for conformance, representation fidelity, validation, achieved performance, readiness, and production authorization.	042

5 Required output-role interpretation

The fixed schema is closed. N_p , C_p , and Q_p are distinct primary accumulators. m_p is derived only on accepted local support. $T_p^{(\kappa^2)}$ is method-specific temporal accounting. No role may be renamed or reinterpreted so that Q_p becomes precision, C_p becomes exposure, coefficient-squared time becomes acquired time, or local map support becomes a standalone availability product.

Whole-bundle and local-pixel outcomes must remain separate. A required whole-role formation or publication failure suppresses the bundle. A local zero, cancellation, or inadequate-support state remains content in a complete map role. A completion marker cannot override a missing required role.

6 Claim discipline

The following conclusions require separate evidence and separate authority:

Claim	Required basis
Algebraic scientific conformance	Complete requirement and prediction assessment under the exact contract identity.
Representation fidelity	Evidence that persisted or transmitted identities, units, states, and roles preserve the conformant scientific objects.
Numerical validation	Authorized profiles establishing realized error and conditioning adequacy; contract text alone is insufficient.
Response/covariance fidelity	Separately authorized products and validation; base v0.1 supplies neither.
Observational performance	Named observational evidence and acceptance authority.
Readiness and production	Explicit operational qualification and owner authorization.

At document revision r0.1, every layer beyond draft algebraic specification remains not assessed or not performed.

7 Canonical notation and units

The symbols in Table 3 are normative in both views of this contract. The generic signal unit is denoted by U . The admitted upstream boundary presently names a nonpolarimetric total-intensity-equivalent quantity in mJy/beam ; the symbol U keeps the algebra explicit and does not turn a component token into a physical Stokes assertion.

Table 3: Canonical notation.

Symbol	Meaning and domain	Unit
a	Stable TolTEC array identity, exactly one of a1100 , a1400 , or a2000 .	identity
i	Exact admitted PTC occurrence on stable RTC output sample n , with observation, detector occurrence/UID, PTC application generation, segment, array, and immutable parents.	identity
p	Target pixel in one exact JINC destination WCS.	identity
z_i	Exact finite transformed signal Z_i^{PTC} .	U
ω_i	Positive, finite, PTC-produced analysis/gridding coefficient from one exact registered family/version that permits SCI-JINC.	1 or U^{-2} , as declared by its family
κ_{ip}	Signed, dimensionless analytic JINC coefficient at the selected point phase. Finite positive, exact-zero, and finite negative values are distinct legitimate analytic results.	1
w_{ip}	Complete signed contribution coefficient, $w_{ip} = \omega_i \kappa_{ip}$.	unit of ω_i
I_{ip}	Boolean membership after sample admission, exact AST association, rounded-center domain admission, local square geometry, and all pixel-local gates.	1
s_a	Explicit angular radial scale associated with stable array a ; finite and strictly positive in any numerical realization.	angle
r	Angular separation between the realized sample phase and destination pixel evaluation point in the target WCS.	angle
r'_a	Dimensionless radius r/s_a , finite and nonnegative.	1
$a_a, b_a, c_a, (r_{\max})_a$	Ordered, dimensionless analytic shape tuple for array a ; every member is finite and strictly positive in an authorized numerical realization.	1
θ_a	Complete kernel parameter identity $(s_a, a_a, b_a, c_a, (r_{\max})_a)$, including owner, source, version, units, array association, and lifecycle state.	mixed, as above
N_p	Signed signal numerator.	$U[\omega]$
C_p	Signed normalization; it may be finite and negative, but exact zero is never accepted.	$[\omega]$
Q_p	Quadratic accumulator; finite and strictly positive on accepted local support.	$[\omega]$
m_p	Normalized estimate N_p/C_p on accepted local support.	U
D_p	Absolute-term sum $\sum_i I_{ip} \omega_i \kappa_{ip} $, used only as construction state for conditioning.	$[\omega]$
ρ_p	Dimensionless cancellation statistic $ C_p /D_p$.	1
A_{pi}	Fixed-state normalized linear operator row $I_{ip} \omega_i \kappa_{ip}/C_p$.	1
$f_{s,i}$	Effective processed-sample frequency, finite and strictly positive.	s^{-1}
$T_p^{(\kappa^2)}$	Required <code>jinc_coefficient_squared_time</code> accumulator.	s

In-memory indices are zero-based. Persisted FITS/WCS pixel coordinates are one-based. Memory order never supplies axis sign, handedness, orientation, or wrapping.

8 Normative definitions and boundary

SCI-JINC observation estimator.

A signed spatial-kernel transformer that consumes exact PTC signal occurrences and their exact AST coordinates and forms at most one complete JINC bundle for each admitted and requested stable array in one observation. It is a sibling alternative to ordinary SCI-MAP, not a downstream stage of SCI-MAP.

Exact occurrence.

The scientific identity that jointly fixes observation, detector occurrence and UID, stable RTC output sample, PTC product and application generation, PTC segment, stable array/group, time attribute, and immutable parent chain. Row position, nearest time, tolerance, shape, detector order, or numerical equality is not a substitute.

Exact AST association.

The relation binding z_i to the AST-owned `SCI-AST:rtc_output_grid_coordinates@1` continuous coordinate for the same processed sample realization in the exact target JINC WCS. Its data-model join is an engineering choice; its scientific identity is not.

JINC sample admission.

The decision of the immutable `SCI-JINC:jinc_map_contribution@1` profile that one exact upstream occurrence may be considered by JINC. It neither implies rounded-center domain admission nor pixel support.

Rounded-center domain admission.

The JINC-owned occurrence-level gate applied after center rounding and before any footprint evaluation. The resolved rounded center must lie within the finite destination pixel domain. An outside center makes $I_{ip} = 0$ for every p , even when the square footprint would overlap the map.

Point phase.

The single-valued residual between the exact continuous sample coordinate and the resolved rounded center, quantized on each axis to select one point-evaluated kernel matrix. It is not a pixel-area average.

Square support.

The fully populated square cache whose half-width is fixed by $(r_{\max})_a$. Every resolved square pixel is evaluated, including corners with radius larger than $(r_{\max})_a$; there is no circular membership predicate.

Sample-pixel membership.

The Boolean proposition $I_{ip} = 1$ after sample admission, exact coordinate association, rounded-center domain admission, square placement, finite local geometry, and every pixel-local precondition. A contract-defined analytic zero is a member with zero contribution; outside support is nonmembership; a finite negative coefficient is an ordinary signed member.

Local formal support.

The state in which the required inputs and accumulators are finite, $Q_p > 0$, $C_p \neq 0$, and cancellation is resolved with total numerical error negligible relative to the approximately 10^{-3} instrument-fidelity scale. A named empirical policy may narrow this state but never promote it.

Complete observation bundle.

The atomic fixed-role product for one observation, stable array, exact JINC plan and realization, target WCS, and lifecycle generation. It contains exactly N_p , C_p , Q_p , derived m_p with local support/validity, and $T_p^{(\kappa^2)}$.

Whole-bundle failure.

Failure to form or publish any required whole-product role. It suppresses the affected bundle without a partial role, placeholder, synthetic empty-array product, or completion override. Pixel-local invalid support is ordinary bundle content and is not whole-bundle failure.

Typed numerical unavailability.

The fail-closed state in which a numerical JINC route cannot be realized because an exact authorized TolTEC kernel parameter set or an exact registered, selected, realized, JINC-permitted PTC coefficient family/payload is absent. It is not repaired by a hidden default, unity coefficient, alternate family, ordinary MAP permission, inherited scale, or implementation value.

Scientific destination identity.

The product identity formed from observation, stable array, exact JINC plan and realization, target WCS, product role, and lifecycle generation. Worker, thread, process, container, filename, vector position, and network order are

not scientific identity.

9 Canonical equations

9.1 Peak-normalized analytic family

Let J_1 be the first-order Bessel function of the first kind. Define the collision-free peak-normalized function

$$\mathcal{J}(x) = \begin{cases} 2J_1(x)/x, & x \neq 0, \\ 1, & x = 0. \end{cases} \quad (1)$$

For $r'_a = r/s_a$ and $j_{1,1} = 3.831706$, the generic SCI-JINC coefficient is

$$\kappa_a(r'_a; a_a, b_a, c_a, (r_{\max})_a) = \mathcal{J}\left(\frac{2\pi r'_a}{a_a}\right) \exp\left[-\left(\frac{2r'_a}{b_a}\right)^{c_a}\right] \mathcal{J}\left(\frac{j_{1,1} r'_a}{(r_{\max})_a}\right). \quad (2)$$

Equation (2) has $\kappa_a(0) = 1$. a_a scales the first JINC argument, b_a scales the generalized-exponential envelope, c_a is its exponent, and $(r_{\max})_a$ places the first zero of the second JINC factor. Analytic zeros and finite signed lobes are retained. The square-cache rule, rather than a radial cutoff, determines membership.

9.2 Signed estimator

For the exact occurrence population of one bundle,

$$w_{ip} = \omega_i \kappa_{ip}, \quad (3)$$

$$N_p = \sum_i I_{ip} \omega_i \kappa_{ip} z_i, \quad (4)$$

$$C_p = \sum_i I_{ip} \omega_i \kappa_{ip}, \quad (5)$$

$$Q_p = \sum_i I_{ip} \omega_i \kappa_{ip}^2, \quad (6)$$

$$m_p = \frac{N_p}{C_p}, \quad (7)$$

$$A_{pi} = \frac{I_{ip} \omega_i \kappa_{ip}}{C_p}. \quad (8)$$

All coupled accumulator terms for one contribution use the same I_{ip} and the same realized κ_{ip} identity. The normalization is signed. A finite negative C_p is admissible; exact $C_p = 0$ is not a zero-sky result.

9.3 Conditioning and local support

Define construction-state quantities

$$D_p = \sum_i I_{ip} |\omega_i \kappa_{ip}|, \quad (9)$$

$$\rho_p = \frac{|C_p|}{D_p}, \quad D_p > 0. \quad (10)$$

For exact arithmetic, $0 \leq \rho_p \leq 1$. Accepted local support requires finite contributors, finite N_p , C_p , and Q_p , $Q_p > 0$, $C_p \neq 0$, and a demonstration that total numerical error is negligible compared with the approximately 10^{-3} relative fidelity relevant to the instrument. Neither D_p nor an error estimate is a persistent base-v0.1 product.

9.4 Coefficient-squared time

The sole authorized base-v0.1 time-support product is

$$T_p^{(\kappa^2)} = \sum_i I_{ip} \frac{\kappa_{ip}^2}{f_{s,i}}. \quad (11)$$

It is nonnegative and has unit seconds. The squared object is κ_{ip} , not ω_i or w_{ip} .

9.5 Conditional mathematics retained for future science

The following identities do not create base-v0.1 products. If and only if an exact PTC family establishes $\omega_i = \text{Var}(z_i)^{-1}$ for mutually independent admitted occurrences,

$$\text{Var}_{\text{formal}}(m_p) = \frac{Q_p}{C_p^2}, \quad (12)$$

$$W_{\text{formal},p} = \frac{C_p^2}{Q_p}, \quad (13)$$

$$\text{Cov}(m_p, m_{p'}) = \frac{\sum_i I_{ip} I_{ip'} \omega_i \kappa_{ip} \kappa_{ip'}}{C_p C_{p'}}. \quad (14)$$

For a general available upstream covariance,

$$\mathbf{C}_{\text{JINC}} = \mathbf{A}_{\text{JINC}} \mathbf{C}_{\text{PTC}} \mathbf{A}_{\text{JINC}}^T. \quad (15)$$

Inverse-square units alone do not establish inverse variance, independence, precision, covariance authority, or significance. Response, variance, formal-weight, and covariance companions remain deferred until a concrete scientific use separately authorizes them.

10 Assumptions, validity classes, and explicit unavailability

Table 4: Normative assumptions and boundary states.

ID	Statement
SCI-JINC-ASM-001	The contract concerns one observation at a time and a nonpolarimetric total-intensity-equivalent transformed signal. It does not authorize a physical Stokes interpretation from a component token.
SCI-JINC-ASM-002	PTC and AST meanings, identities, validity facts, causes, uncertainty, and immutable ancestry are producer-owned facts. JINC uses them without reconstructing or redefining them.
SCI-JINC-ASM-003	An admitted ω_i is finite and strictly positive and comes from one exact PTC family/version that explicitly permits SCI-JINC. No registered family or realized payload is supplied by the scientific author packet.
SCI-JINC-ASM-004	Equations (12)–(15) are conditional identities only. Their assumptions are not inferred from units, names, or another consumer’s permission.
SCI-JINC-ASM-005	The analytic family and parameter meanings are approved, but no TolTEC numerical realization of s_a , a_a , b_a , c_a , $(r_{\max})_a$, pixel size, or <code>subpixel_n</code> is scientifically supplied or recommended by this contract.
SCI-JINC-ASM-006	Point-phase evaluation and fully populated square support are the scientific operator. Pixel-area integration, a circular cutoff, footprint-overlap admission, and JINC-then-crop equivalence are not alternative branches.

ID	Statement
SCI-JINC-ASM-007	Total numerical error includes finite arithmetic, accumulation/reduction order, analytic-function evaluation, phase quantization, and cache/index realization. Adequacy is judged against the approximately 10^{-3} relative instrument-fidelity scale; no stronger universal precision claim follows.
SCI-JINC-ASM-008	Same-observation chunking may be an engineering realization of one bundle, but a chunk is not a scientific product identity. Cross-observation combination is outside base v0.1.
SCI-JINC-ASM-009	This Stage B draft has inspected no implementation candidate, configuration, product, test, audit, validation, reduction, or production evidence.

Every presently missing fact is classified exactly once in Table 5.

Table 5: Missing-fact classification.

Class	Fact	Consequence
Approved	Generic analytic identity, signed estimator, point-phase square operator, center-domain gate, conditioning principle, fixed bundle, grouping, and ownership boundary.	These are normative in v0.1.
Known but not supplied	A registered, selected, and realized JINC-permitted PTC coefficient family and payload.	No numerical bundle is formed without it; no unity or alternate-family fallback.
Known but not supplied	A scientifically authorized TolTEC array-associated numerical parameter set.	No numerical bundle is formed; no inherited scale, shape, extent, or mode-dependent value is used.
Unresolved decision	None within the approved base-v0.1 scientific scope.	No owner question is raised by this draft.
Implementation unassessed	Representation, interfaces, algorithms, storage, parallel realization, and implementation behavior.	No conformity or representation-fidelity claim.
Validation not performed	Algebraic execution, numerical adequacy, observational response, covariance, photometry, performance, readiness, and production operation.	No achieved result, readiness, or production claim.

Response, covariance, formal weight, physical exposure, cross-observation coaddition, standalone support or availability, diagnostics, and generalized provenance are deferred or outside base v0.1, rather than silently missing members of the fixed bundle.

11 Binding scientific requirements

The key words *shall*, *shall not*, *required*, and *unavailable* are normative. These requirements define scientific conformance; they do not assert that any implementation satisfies them.

SCI-JINC-REQ-001. SCI-JINC shall define a signed spatial-kernel estimator for one observation and stable array. It shall consume exact PTC and AST parents directly, consume no SCI-MAP product, and inherit no ordinary SCI-MAP normalization, support, exposure, validity, response, covariance, product, or coaddition rule by analogy.

SCI-JINC-REQ-002. Every candidate contribution shall bind one exact occurrence and the AST continuous coordinate associated with the same processed sample realization in the exact target JINC WCS. Row order, detector

order, nearest time, tolerance, shape, cardinality, or numerical coordinate equality shall not establish that association.

SCI-JINC-REQ-003. The transformed signal z_i shall retain its exact PTC quantity, unit, availability, application generation, retention state, immutable parents, and producer causes. SCI-JINC shall spatially transform that quantity without creating calibration, conditioning, astrometry, empirical noise, or source interpretation.

SCI-JINC-REQ-004. Every ω_i shall come from one exact family/version in the single PTC-owned positive analysis/gridding coefficient registry. The family shall explicitly permit SCI-JINC, and its requested, effective, observation-resolved, and realized identities, payload, QC, normalization, support, provenance, covariance meaning, uncertainty, and prohibited meanings shall remain bound.

SCI-JINC-REQ-005. A numerical route shall fail closed when coefficient selection is missing without an authorized versioned mode default, the family is unregistered, JINC permission is absent, or the payload/QC/identity is absent, unavailable, incompatible, duplicate, or ambiguous. SCI-JINC shall not infer unity, another family, ordinary MAP permission, **sens**, loading, scatter, inverse variance, precision, or significance.

SCI-JINC-REQ-006. After the producer-owned coefficient decision, SCI-JINC shall admit only a finite strictly positive ω_i . Zero, negative, non-finite, or unrepresentable upstream coefficient values shall not be deposited and shall retain the exact profile-defined disposition and cause.

SCI-JINC-REQ-007. Every numerical kernel route shall bind one exact scientifically authorized array-associated parameter-set identity containing finite positive s_a , a_a , b_a , c_a , $(r_{\max})_a$, finite positive pixel size, and effective integer `subpixel_n` ≥ 1 . Missing authority shall make the affected route unavailable without fallback.

SCI-JINC-REQ-008. The peak-normalized analytic primitive shall be $\mathcal{J}(x) = 2J_1(x)/x$ for $x \neq 0$ with the removable limit $\mathcal{J}(0) = 1$, as in Equation (1).

SCI-JINC-REQ-009. The analytic coefficient shall follow Equation (2), including its fixed parameter ordering and $r'_a = r/s_a$. It shall have peak one, preserve analytic zeros, and preserve every finite positive and negative lobe of the complete product.

SCI-JINC-REQ-010. The sample center shall be resolved by a single-valued rounding rule, its two-axis residual phase shall be quantized, and one point-evaluated kernel matrix shall be selected. This operator shall not be represented as or replaced by pixel-area integration.

SCI-JINC-REQ-011. Tie handling, phase intervals, bin edges, representatives, cache-index rounding, analytic-function evaluation, and accumulation order may be engineering choices only when they are single-valued, preserve the accepted point-phase operator, and meet the numerical-adequacy requirement.

SCI-JINC-REQ-012. $(r_{\max})_a$ shall place the first zero of the second JINC factor and fix the fully populated square-cache half-width. Every resolved square pixel, including a corner with radius greater than $(r_{\max})_a$, shall be evaluated; no circular cutoff or radial membership predicate shall be applied.

SCI-JINC-REQ-013. Before any footprint evaluation, the resolved rounded cache center shall lie within the finite destination pixel domain. An outside center shall set $I_{ip} = 0$ for every pixel, even when its square would overlap the map; footprint-overlap admission is prohibited.

SCI-JINC-REQ-014. For an admitted in-map center, square pixels outside the finite map shall be absent. The route shall not wrap, reflect, complete the footprint, substitute an interior normalizer, renormalize the edge, or require JINC-then-crop equivalence.

SCI-JINC-REQ-015. Every evaluated κ_{ip} shall be finite. A finite negative coefficient is normal and shall retain its sign; an analytic zero is a contract-defined zero contribution distinct from outside support; a non-finite coefficient shall fail the affected required bundle before deposition rather than be clipped, skipped, or zero-filled.

SCI-JINC-REQ-016. All coupled accumulator terms for one contribution shall use the same exact occurrence, AST association, sample-admission decision, rounded-center decision, sample-pixel membership, phase, and κ_{ip} identity.

SCI-JINC-REQ-017. The required accumulators shall be distinct and shall obey Equations (4)–(6). Negative lobes shall keep their sign in N_p and C_p and enter Q_p quadratically.

SCI-JINC-REQ-018. The published signal estimate shall be $m_p = N_p/C_p$ and the fixed-state normalized operator shall be Equation (8). No Q_p denominator or other local measurement model shall be substituted.

SCI-JINC-REQ-019. Local formal support shall require finite admitted inputs, finite N_p , C_p , and Q_p , $Q_p > 0$, $C_p \neq 0$, and numerically adequate resolved cancellation. A rejected pixel shall carry local invalid support in the `jinc_map` role and shall not receive a finite substitute.

SCI-JINC-REQ-020. A finite negative C_p shall be admissible. Exact $C_p = 0$ shall be locally invalid and unavailable, never interpreted as zero sky, infinity, or a valid non-finite map value.

SCI-JINC-REQ-021. SCI-JINC shall retain the dimensionless conditioning statistic ρ_p of Equation (10). A finite nonzero result shall be supported only when total numerical error is demonstrably negligible relative to the approximately 10^{-3} instrument-fidelity scale; otherwise near cancellation shall be locally invalid.

SCI-JINC-REQ-022. No universal ρ_p cutoff, unit-bearing C_p or Q_p floor, contributor-count/machine-epsilon formula, prescribed summation algorithm, fixed reduction order, bitwise reproducibility, or stronger precision shall be treated as scientific authority.

SCI-JINC-REQ-023. Total numerical error from finite arithmetic, accumulation and reduction order, analytic-function evaluation, phase quantization, and cache/index realization shall remain negligible relative to the approximately 10^{-3} instrument-fidelity scale while preserving exact membership and the point-phase square operator.

SCI-JINC-REQ-024. The unit relations in Table 3 and the two allowed ω_i unit classes shall be preserved. Common positive scaling of all ω_i shall leave m_p , ρ_p , and A_{pi} invariant; signal-unit rescaling shall transform the estimator consistently.

SCI-JINC-REQ-025. Equations (12)–(15) may receive a formal statistical interpretation only when the exact PTC family and covariance assumptions establish it. Inverse-square units or historical labels alone shall not establish variance, precision, covariance, or significance.

SCI-JINC-REQ-026. The required time-support product shall be exactly Equation (11). It shall square κ_{ip} , not ω_i or w_{ip} , and shall have unit seconds.

SCI-JINC-REQ-027. `jinc_coefficient_squared_time` shall be described only as method-specific coefficient-squared temporal accounting. It shall not be interpreted as physical acquired exposure, valid-original exposure, complete temporal support, hits, normalized influence, white-noise-equivalent time, precision, validity, confidence, or significance.

SCI-JINC-REQ-028. Every produced base-v0.1 bundle shall contain exactly five required roles: `jinc_signal_numerator`, `jinc_signed_normalization`, `jinc_quadratic_accumulator`, derived `jinc_map` with local support/validity, and `jinc_coefficient_squared_time`.

SCI-JINC-REQ-029. Failure to form or publish any required whole-product role shall suppress the affected bundle atomically. Pixel-local zero, insufficient, cancelled, or otherwise invalid support shall remain ordinary bundle content and shall not suppress an otherwise complete role or create a role-availability object.

SCI-JINC-REQ-030. One observation may produce zero through three independent bundles, with at most one for each stable array admitted and requested under the exact JINC realization. A missing, unavailable, or unrequested array shall create no placeholder or empty-array product and shall not invalidate another array's bundle.

SCI-JINC-REQ-031. Each bundle shall be independently identified by observation, stable array, exact JINC plan and realization, target WCS, product role, and lifecycle generation. Contributions with different array or destination identities shall never merge.

SCI-JINC-REQ-032. The complete destination identity shall be resolved before allocation, numerical mutation, output side effects, or parallel execution. Ambiguous, duplicate, or aliased destination ownership shall fail the affected bundle without selecting a winner or publishing a partial result.

SCI-JINC-REQ-033. Same-observation samples or chunks may accumulate incrementally only when observation, stable array, JINC plan and realization, target WCS, admission/parameter/coefficient state, destination, and lifecycle generation are identical. Chunk, worker, process, container, filename, and vector position shall not create product identity.

SCI-JINC-REQ-034. Base v0.1 shall authorize no cross-observation combination. Any future JINC coadd shall require a separately authorized boundary over complete observation bundles and shall not infer accumulator addition, normalized-map combination, or ordinary SCI-MAP coaddition.

SCI-JINC-REQ-035. The immutable `SCI-JINC:jinc_map_contribution@1` evaluation shall remain distinct from PTC signal availability, PTC retention, coefficient availability/QC, AST coordinate validity and association, finite signal, finite positive ω_i , parameter availability, rounded-center admission, pixel support, finite κ_{ip} , cancellation, and final bundle validity.

SCI-JINC-REQ-036. Only a fact named by an exact restriction shall control JINC admission. Producer causes and transitive influence shall be preserved, but a generic flag or cause shall have no universal veto. Outside support, an analytic zero, or an outside rounded center shall be ordinary no-contribution; a finite negative lobe shall not create a cause.

SCI-JINC-REQ-037. ALIGN/AST, RTC, CAL, PTC, and VAL shall retain the meanings they own. SCI-JINC shall own only its admission use, local geometry, point phase, square support, signed coefficient, normalization, conditioning, fixed accumulators, local support, grouping, destination, and atomic bundle publication.

SCI-JINC-REQ-038. Base v0.1 shall publish no response, variance, formal-weight, covariance, empirical-noise, or significance product. The conditional mathematics in Section 9.5 shall remain preserved for a future concrete scientific use without being promoted to a present role.

SCI-JINC-REQ-039. Base v0.1 shall publish no physical-exposure product. Any future exposure role shall require separate authority for exact original- occurrence lineage and semantics and shall not reinterpret $T_p^{(\kappa^2)}$.

SCI-JINC-REQ-040. Base v0.1 shall create no standalone support or availability role, generic optional or conditionally required role mechanism, detailed operational-cause archive, diagnostic role, generalized provenance product, or per-pixel or per contribution provenance system.

SCI-JINC-REQ-041. No TolTEC numerical parameter realization, recommendation, optimum, hidden default, or fallback is authorized by v0.1. In particular, an inherited 45 m denominator, inherited shape values, or mode-dependent $r_{\max}$ values shall not be treated as scientific authority.

SCI-JINC-REQ-042. A future conformance assessment shall exercise every prediction in Section 12, disclose its numerical acceptance policy, and separate algebraic correctness from implementation conformity, representation/response fidelity, numerical and observational validation, achieved performance, readiness, and production authorization. This Stage B draft establishes none of those later claims.

12 Falsifiable predictions and edge cases

These predictions are normative tests for a future authorized conformance program. Listing them is not evidence that they have been executed or passed.

Table 6: Stable SCI-JINC predictions.

Prediction	Falsifiable expected result
SCI-JINC-PRED-001	A constant finite input $z_i = z_0$ returns $m_p = z_0$ on every accepted local support, including a cropped edge pixel.
SCI-JINC-PRED-002	One admitted contributor with nonzero finite κ_{ip} returns $m_p = z_i$, whether the coefficient is positive or negative.
SCI-JINC-PRED-003	A hand-computed two-contributor case with unequal z_i , ω_i , and signed κ_{ip} matches the separate N_p , C_p , Q_p , m_p , and $T_p^{(\kappa^2)}$ equations.
SCI-JINC-PRED-004	Multiplying every ω_i by the same positive factor scales N_p , C_p , and Q_p together and leaves m_p , ρ_p , and A_{pi} unchanged.
SCI-JINC-PRED-005	Under a true inverse-variance family, rescaling $z_i \mapsto \lambda z_i$ and $\omega_i \mapsto \omega_i/\lambda^2$ produces $m_p \mapsto \lambda m_p$ and conditional formal weight $W_{\text{formal},p} \mapsto W_{\text{formal},p}/\lambda^2$.
SCI-JINC-PRED-006	The analytic evaluator returns $\kappa_a(0) = 1$ through the removable limit, not a non-finite value or an implementation-dependent substitute.
SCI-JINC-PRED-007	An analytic $\kappa_{ip} = 0$ changes none of N_p , C_p , Q_p , or $T_p^{(\kappa^2)}$ while remaining distinguishable from outside square support.
SCI-JINC-PRED-008	A finite negative lobe retains its sign in N_p and C_p , enters Q_p and $T_p^{(\kappa^2)}$ quadratically, and is not clipped, absolutized, omitted, or flagged solely for being negative.
SCI-JINC-PRED-009	Exact cancellation $C_p = 0$ yields local invalid support and no finite substitute, even when N_p and Q_p are finite.
SCI-JINC-PRED-010	A finite negative C_p with otherwise adequate support yields the signed normalized estimate and is not rejected merely for the denominator sign.
SCI-JINC-PRED-011	A near-cancellation construction whose numerical error is demonstrated negligible relative to the 10^{-3} scale remains eligible; no universal ρ_p cutoff rejects it.
SCI-JINC-PRED-012	A near-cancellation construction lacking that adequacy demonstration is locally invalid, even when finite arithmetic happens to return a nonzero C_p .
SCI-JINC-PRED-013	A non-finite required signal, geometry, coefficient evaluation, accumulator, or sample frequency cannot be masked by multiplication by zero or silently skipped; the exact local or whole-bundle failure rule applies.
SCI-JINC-PRED-014	Square-cache corners whose radius exceeds $(r_{\text{max}})_a$ are evaluated by Equation (2) and may contribute.
SCI-JINC-PRED-015	Two points on opposite sides of a hypothetical circular r_{max} boundary but inside the same resolved square are both governed by square membership; no radial cutoff appears.
SCI-JINC-PRED-016	Effective <code>subpixel_n=1</code> selects one single-valued point-phase matrix and does not become a pixel-area average.
SCI-JINC-PRED-017	Coordinates immediately around every half-pixel tie and phase-bin edge select a single-valued realization whose total error meets the numerical-adequacy requirement; no bitwise cross-realization identity is presumed.
SCI-JINC-PRED-018	A rounded center immediately outside each map edge or corner changes none of the four accumulators for any pixel, even when its square footprint overlaps the map.
SCI-JINC-PRED-019	A rounded center immediately inside an edge uses only retained square pixels with no wrap, reflection, completion, renormalization, or edge correction.
SCI-JINC-PRED-020	A supported constant field at an in-map cropped edge still returns the constant through the actual retained N_p/C_p terms.
SCI-JINC-PRED-021	$T_p^{(\kappa^2)}$ is unchanged by changing ω_i alone, receives zero from an analytic zero, and receives a positive term from either sign of nonzero κ_{ip} .

Prediction	Falsifiable expected result
SCI-JINC-PRED-022	Missing coefficient selection without an authorized default, an unregistered family, missing JINC permission, or an unavailable/mismatched payload prevents the affected numerical bundle without unity or alternate-family fallback.
SCI-JINC-PRED-023	Absence of an authorized TolTEC parameter set prevents the numerical bundle; no inherited 45 m denominator, shape values, mode-dependent $r_{\max}$, or other implementation value is substituted.
SCI-JINC-PRED-024	Permuting arrays, networks, detector rows, map groups, or container positions does not change array-to-parameter or array-to-bundle identity; a missing array creates no placeholder and does not invalidate another array.
SCI-JINC-PRED-025	Ambiguous, duplicate, or aliased destination ownership fails the affected bundle before mutation and produces no partial winner.
SCI-JINC-PRED-026	Valid decompositions of the same observation into chunks select the same scientific memberships and equations and agree within the disclosed numerical-adequacy policy; bitwise equality is not required.
SCI-JINC-PRED-027	An attempted cross-observation accumulator or map combination is rejected as outside base v0.1 rather than being assigned an inferred coadd rule.
SCI-JINC-PRED-028	A missing, duplicate, or ambiguous exact signal-coordinate association prevents the affected contribution and is not reclassified as ordinary outside support.
SCI-JINC-PRED-029	Failure to form any one of the five required whole-product roles suppresses the bundle, whereas a locally invalid pixel does not suppress an otherwise complete role.
SCI-JINC-PRED-030	The base bundle contains no response, covariance, formal-weight, physical-exposure, standalone availability/support, diagnostic, generalized provenance, or placeholder product.
SCI-JINC-PRED-031	With dimensionless ω_i , Q_p/C_p^2 remains dimensionless and is not reported as signal variance or used to create a formal-weight product.
SCI-JINC-PRED-032	The JINC route consumes no SCI-MAP product and no ordinary MAP admission, validity, exposure, support, response, covariance, or coaddition decision is substituted for a JINC-owned decision.

12.1 Prediction-to-requirement trace

Predictions	Governing requirements
001-003	016-020, 026
004-005	024-025
006-008	008-009, 015, 017, 026
009-012	019-023
013	003, 006-007, 015, 019, 026, 029
014-017	010-012, 023
018-020	013-014, 016-020
021	026-027
022-023	004-007, 041
024-027	030-034
028	002, 035-036
029-030	028-029, 038-040
031	024-025, 038
032	001, 035-038

13 Blank conformance-result template

The following fields are intentionally blank until a separately authorized assessment binds a candidate. Their presence creates no assessment claim.

Field	Future assessment value
Candidate identity	Not assessed
Contract and source binding	Not assessed
Coefficient family and permission	Not assessed
TolTEC parameter-set identity	Not assessed
Observation, array, WCS, and destination	Not assessed
Requirement results	Not assessed
Prediction results	Not assessed
Numerical acceptance policy	Not assessed
Representation-fidelity result	Not assessed
Validation result	Not performed
Achieved-performance result	No claim
Readiness/production authorization	No claim